Tight Blood Glucose Control without Early Parenteral Nutrition in The ICU

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Abstract

Background: Previous randomized controlled trials (RCTs) have shown benefit and harm from tight blood glucose control (tight-GC) in the intensive care unit (ICU). Variation in early parenteral nutrition use and in insulin-induced severe hypoglycemia might explain this inconsistency.

Methods: Upon ICU admission, patients in this study were randomized to liberal-GC tolerating hyperglycemia up to 215 mg/dL (11.9 mmol/L), or tight-GC (80-110 mg/dL; 4.4-6.1 mmol/L) using the LOGIC-Insulin algorithm, while withholding parenteral nutrition for one week. Protocol adherence was reported by glucose metrics. Time to discharge alive from ICU, accounting for death as competing risk, was the primary end point; 90-day mortality was the safety end point. Heterogeneity in treatment effects for six predefined subgroups was analyzed. Impact on eight secondary outcomes was reported by point-estimates and confidence intervals.

Results: Of 9230 randomized patients, 4622 were assigned to liberal-GC and 4608 to tight-GC. Average morning blood glucose (median [P25-P75]) was 140(122-161) mg/dL with liberal-GC and 107(98-117) mg/dL with tight-GC. Severe hypoglycemia occurred in 31 (0.7%) of liberal-GC patients and 47 (1.0%) of tight-GC patients. Time to discharge alive from ICU was similar for both groups (Hazard Ratio [95% CI] 1.00[0.96-1.04], P=0.94). Ninety-day mortality was also comparable (10.1% with liberal-GC; 10.5% with tight-GC, P=0.51) with possible treatment heterogeneity for neurological/neurosurgical patients. Severe acute kidney injury, new use of renal replacement therapy and cholestatic liver dysfunction were less prevalent with tight-GC.

Conclusion: In critically ill patients not receiving early parenteral nutrition, tight-GC did not alter time to discharge alive from ICU or mortality.

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In critically ill patients admitted to the intensive care unit (ICU), hyperglycemia is common and is associated with poor outcome.[1-3] Whether this association reflects causality remains debated, as randomized controlled trials (RCTs) have shown divergent effects of blood glucose lowering.[4-11] In three single-center RCTs, lowering blood glucose with insulin to the healthy, age-adjusted fasting range (tight-GC) improved outcome as compared with tolerating hyperglycemia.[4-6] Mechanistic studies attributed this benefit to prevention of glucose toxicity in cells such as neurons, renal tubular cells, hepatocytes and immune cells that take up glucose in an insulin-independent, glucose concentration gradient-dependent manner.[12-15] However, benefit of tight-GC was not confirmed in subsequent multicenter RCTs, and the largest trial showed increased mortality,[9] which was attributed to a substantially increased incidence of severe hypoglycemia (<40 mg/dL or 2.2 mmol/L).[16]

These opposite outcome effects might be explained by two important methodological differences.[17] Whereas the trial showing harm from tight-GC has been criticized for unstandardized blood glucose measurements and insulin titration that may have increased the incidence of hypoglycemia, generalizability of the benefit shown in the earlier trials was questioned, as it was observed in a context of early use of parenteral nutrition.[4-6,18] Subsequent RCTs have shown that early parenteral nutrition not only does not improve outcome, it may also increase the risk of infections and delay recovery from critical illness.[19,20] These findings resulted in revised clinical practice guidelines that no longer advocate for early parenteral nutrition.[21] Avoiding early parenteral nutrition also appears to reduce the severity of hyperglycemia and preserves autophagic cell damage removal which may increase the threshold for hyperglycemic toxicity.[19,20,22-24] Moreover, the risk of hypoglycemia with tight-GC may increase in the absence of early parenteral nutrition.[19,20] However, there is evidence that hypoglycemia can be largely avoided with use of the LOGIC-Insulin computer algorithm,

which provides guidance to bedside nurses on how to titrate insulin infusion, as shown in a single center study and subsequently confirmed in a multicenter RCT.[25,26]

Given the evidence, we hypothesized that for critically ill patients not receiving early parenteral nutrition, tight-GC with insulin infusion titrated with use of the LOGIC-Insulin computer algorithm safely reduces intensive care dependency though with a smaller effect size than previously reported for patients receiving early parenteral nutrition.

Methods

Study Design and Oversight

This investigator-initiated trial, with the acronym "TGC-fast", was a prospective, randomized, controlled, parallel-group multicenter study. The study protocol and informed consent forms were approved by the ethics committees of the participating centers and by the Belgian authorities. A description of the study protocol and statistical analysis plan has been published and is available at NEJM.org.[27] Written informed consent was obtained from each patient or a legal representative.

Patients

From September 2018 until August 2022, consecutive adult patients admitted to one of eleven ICUs from two university hospitals and one district hospital in Belgium were screened for eligibility (Fig. 1). During the first 2020 Covid-19 wave, recruitment was temporarily halted as mandated by the central ethics committee. Upon resumption of recruitment, patients admitted to closed Covid-19 units were no longer assessed for eligibility. Eligible patients were allocated 1:1 to liberal-GC or tight-GC by centralized computer randomization that used permuted blocks of 10 and was stratified by center and diagnostic category upon ICU admission. Bedside nurses and physicians were unaware of the block size.

Study Procedures

For patients in the tight-GC group, blood glucose concentration was targeted at 80 to 110 mg/dL (4.4 to 6.1 mmol/L). The insulin doses, glucose doses in case of hypoglycemia, and blood glucose measurement frequency (ranging from every 1 to 4 hours and more frequently after hypoglycemia) were noted by the LOGIC-Insulin computer algorithm.[25,26]

For patients in the liberal-GC group, insulin was only initiated when blood glucose exceeded 215 mg/dL (11.9 mmol/L) on two consecutive measurements (or 1 measurement for patients with type-1 diabetes), after which insulin was titrated by bedside physicians and nurses to target blood glucose between 180 and 215 mg/dL (10 and 11.9 mmol/L). To optimize protocol adherence, a simple alert was integrated in the patient data management systems of each center to inform physicians and nurses on when to administer or stop insulin. Blood glucose concentration was measured at least 4 times daily. Management of hypoglycemia was at the discretion of the attending physician in each ICU at the time.

In both groups, blood glucose concentrations were measured on arterial blood with use of a blood gas analyzer, and insulin was only administered as a continuous intravenous infusion through a central venous catheter. When the arterial line was removed, capillary blood could be used for blood glucose measurements. The study intervention was stopped when the patient started to eat, was no longer in need of a central venous line, or was discharged from the ICU, whichever came first. In patients readmitted to the ICU within 48 hours after discharge, the initially allocated randomized intervention was resumed. All patients received enteral nutrition as soon as possible. When enteral nutrition was insufficient to meet the caloric target, parenteral nutrition was only initiated after one week in the ICU. For all patients who did not receive 80% of nutritional intake enterally, parenteral micronutrients were administered as per local practice, to prevent refeeding syndrome.

Outcomes

The primary endpoint was the duration of intensive care dependency, which was taken as the time to discharge alive from ICU to account for death as a competing risk. This was done with use of “time to ready for discharge from ICU,” defined as the time at which patients were no

longer at risk or in need of vital organ support or the time they were actually discharged, whichever came first. The safety endpoint was mortality at 90 days after randomization. The incidence of severe hypoglycemia (<40 mg/dL; <2.2 mmol/L) that was resistant to intravenous glucose administration was considered a serious adverse event.

Eight secondary outcomes were predefined. A first was ICU-acquired kidney injury, defined by incidence of severe acute kidney injury (AKI-3) according to the creatinine criteria of the Kidney Disease Improving Global Outcome (KDIGO) guidelines[28] and by new use of renal replacement therapy. Another secondary outcome was liver dysfunction as revealed by cholestatic or cytolytic liver dysfunction markers (gamma glutamyltransferase ≥90 U/L [1.5 times upper limit of normality], alkaline phosphatase ≥195 U/L [1.5 times upper limit of normality], bilirubin >3 mg/dL, aspartate transaminase ≥111 U/L [3 times upper limit of normality], or alanine transaminase ≥123 U/L [3 times upper limit of normality]).[19] The six other secondary outcomes were new infections, respiratory and hemodynamic support, time to discharge alive from hospital, and ICU and hospital mortality.

Statistical Analysis

We hypothesized much smaller outcome benefits from tight-GC as compared with what we had found in our previous RCTs based on expected smaller differences in blood glucose concentrations between the tight-GC group and the liberal-GC group in the absence of early use of parenteral nutrition. Therefore, the sample size (N=9230) was calculated on the ability to detect a one-day difference in the primary endpoint and a 1.5% difference in mortality to exclude harm (safety endpoint) with 80% power and alpha of 0.05.

Two preplanned interim analyses by the independent data safety monitoring board (DSMB) were performed.[27] The first was done after inclusion of 25% of the preplanned sample size, to assess any need of re-powering based on the real versus estimated duration of ICU stay in the control group. The second was done for the impact on the safety endpoint (mortality) after inclusion of 50% of patients. At both time points, the DSMB allowed continuation of recruitment as planned.

Analyses of differences between randomized treatment groups were performed with use of the JMP®Pro17.0.0 software (SAS Institute Inc, Cary, NC). Data were summarized as frequencies and percentages and medians (25th and 75th percentile) unless indicated otherwise. Mortality was compared using the chi-square test. The cumulative survival and the cumulative incidence of discharge alive from ICU were documented by Kaplan-Meier plots. The time to event analyses were done by Cox-proportional hazards, after assuring, with the Schoenfeld residuals method, that the proportional hazard assumption was met. For time to event analyses, data for non-survivors were censored at a time point beyond that of the last surviving patient to account for death as competing risk.[29]

We investigated potential heterogeneity in treatment effects for six prespecified subgroups. These were patients with a history of diabetes;[3,30] patients after cardiac surgery; patients after (complicated) surgery or trauma; patients with a neurological/neurosurgical admission diagnosis;[30-32] patients with sepsis; and patients with a predicted ICU stay of over a week (through random-forest modeling with use of data available upon ICU admission; for methodological details and performance measures, see Supplementary Appendix and Figures S1-S6 and Tables S1-S2). These predicted long-stay patients, in case of insufficient enteral nutrition, expectedly receive supplemental parenteral nutrition from day 8 onwards, which

increases blood glucose and insulin requirements and may hamper recovery of cell damage.[19,22-24] Interaction between the randomized treatment and the yes/no level for these subgroups was determined with use of Cox-proportional hazard and logistic regression models, followed by visualization of the treatment effects for the subgroups in forest plots.

All analyses were performed on the intention-to-treat population. Two-sided P-values below 0.05 were considered to indicate significance. For secondary outcomes and subgroups, point estimates and 95% confidence intervals of the difference were not adjusted for multiplicity and may not be used in place of hypothesis testing.

Author Contributions

JG and GVdB designed the study protocol, analyzed the data, vouch for the data and the analyses and wrote the paper. Other authors gathered data, approved the decision to publish, and read and provided revisions. More information is given in the online supplement. No confidentiality agreements, except agreed confidentiality before publication, were in place between sponsor and authors/institutions.

Results

Patients and Study Intervention

Of the 24,195 patients assessed for eligibility, 9230 were randomized, 4622 to the liberal-GC group and 4608 to the tight-GC group, all included in the intention-to-treat analysis (Fig.1). Baseline characteristics were comparable for both groups (Table 1). Table S3 describes the representativeness of the study population.

Blood glucose concentrations were comparably elevated upon ICU admission in both groups (Table 1). Thereafter, patients in the tight-GC group had lower blood glucose concentrations and received higher insulin doses than those in the liberal-GC group (Table 2, Fig. 2). In the liberal-GC group, blood glucose concentrations and insulin requirements became substantially higher in the second week in ICU in parallel with the increased nutritional intake from day 8 onwards (Table 2, Fig.2). The incidence of severe hypoglycemia was low (31/4622 [0.7%] in the liberal-GC group and 47/4608 [1.0%] in the tight-GC group) and not different between the two groups (Table 2). There were no patients with therapy-resistant hypoglycemia.

Primary and Safety Endpoints

The time to discharge alive from ICU was not significantly different between groups (Hazard Ratio [HR] 1.00 [0.96-1.04], P = 0.94) (Fig. 2). Within 90 days of randomization, 468/4621 (10.1%) of liberal-GC patients and 486/4607 (10.5%) of tight-GC patients had died (P=0.51) (Fig. 2, Table 2).

Secondary Outcomes

AKI-3 developed in 387/4500 (8.6%) patients in the liberal-GC group and in 326/4507 (7.2%) patients in the tight-GC group; Relative Risk (RR) 0.84 (0.73-0.97). New renal replacement

therapy was initiated in 259/4500 (5.8%) liberal-GC patients and in 212/4507 (4.7%) tight-GC patients; RR 0.82 (0.68-0.98) (Table 2). Biochemical markers of cholestatic liver dysfunction, but not cytolytic liver dysfunction, were lower in the tight-GC than in the liberal-GC group (Table 2). In the liberal-GC group, 1370/4170 (32.9%) of patients developed plasma gamma glutamyltransferase concentrations ≥90 U/L versus 1175/4146 (28.3%) in the tight-GC group; RR 0.86 (0.81-0.92). In the liberal-GC group, 498/3818 (13.0%) of patients had plasma alkaline phosphatase concentrations ≥195 U/L versus 437/3787 (11.5%) in the tight-GC group; RR 0.88 (0.78-0.99). Incidence of new infections, duration of respiratory and hemodynamic support, time to discharge alive from hospital, and ICU and hospital mortality were similar in both groups (Table 2, Table S4).

Analyses of Heterogeneity in Treatment Effects for Primary and Safety Endpoints

For the safety endpoint, 90-day mortality (see Fig. 3), there was interaction between randomization to tight-GC versus liberal-GC and the baseline variable "neurological/neurosurgical admission diagnosis versus other diagnoses" pointing to possible lower mortality with tight-GC in the neurological/neurosurgical subgroup (Fig. 3). No other treatment heterogeneity was observed.

Discussion

In this large multicenter RCT of adult critically ill patients for whom parenteral nutrition was withheld for one week in the ICU, hyperglycemia was much less pronounced as compared with earlier RCTs that used early parenteral nutrition. Further lowering blood glucose into the healthy fasting range with tight-GC, guided by the LOGIC-Insulin algorithm that avoided iatrogenic severe hypoglycemia, did not affect the duration of intensive care dependency or mortality. Secondary morbidity outcomes were also largely unaffected by tight-GC, although the incidence of severe acute kidney injury, the new use of renal replacement therapy and the incidence of cholestatic liver dysfunction were lower in the tight-GC than in the liberal-GC group. These findings suggest that early parenteral nutrition use in previous studies was an important iatrogenic factor that aggravated hyperglycemia into a potentially toxic range.[22,23,36] These results add evidence to the recommendation to omit early parenteral nutrition for adult ICU patients as this reduces need of blood glucose control.

This RCT was performed in a heterogeneous patient population, which allowed assessment of heterogeneity in treatment effect for large subgroups. Possible treatment heterogeneity was limited to patients admitted with a neurological/neurosurgical diagnosis, who revealed lower 90-day mortality with tight-GC than with liberal-GC. A possible explanation could be that the brain is particularly sensitive to harm from hyperglycemia, which may have been aggravated by glucocorticoid therapy, and to harm from hypoglycemia, which in this RCT was effectively prevented by the LOGIC-Insulin algorithm.[37-39] This contrasts with previous RCTs that investigated impact of blood glucose control in stroke patients, which revealed smaller differences in blood glucose between study groups[40] and a substantially higher incidence of iatrogenic hypoglycemia.[32,40]

The limitations of the present trial included the inability to blind caregivers for treatment allocation and the lack of correction for multiple comparisons for secondary outcomes. The primary endpoint, unfortunately, may have been confounded by fluctuating discharge policies due to shortage of ICU beds, a major problem during the Covid-19 pandemic. Also, the clinical relevance of the lower incidence of kidney and liver dysfunction with tight-GC and of the possible treatment heterogeneity for neurological/neurosurgical patients remains uncertain. The strengths comprise the very large sample size and excellent protocol adherence with use of the LOGIC-Insulin algorithm.

In conclusion, in critically ill patients not receiving early parenteral nutrition, hyperglycemia was less severe than reported with parenteral nutrition and further lowering blood glucose into the normal fasting range with the LOGIC-Insulin computer algorithm avoided iatrogenic hypoglycemia without altering mortality or duration of intensive care dependency.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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Figure Legends

Figure 1: Consort diagram with all information on patients who were assessed for eligibility, who were excluded from the trial and who were randomized and used for data analysis.

Figure 2: Panels illustrate blood glucose control, and parenteral and total nutrition in study groups, and show similarity for 90-day survival and for time to discharge alive from ICU. Data points reflect mean values and error bars represent the 95% confidence intervals; the shaded area for Kaplan-Meier survival and cumulative incidence of live ICU discharge plots represent 95% confidence intervals. The widths of the 95% confidence intervals (CI) have not been adjusted for multiplicity and may not be used in place of hypothesis testing. Two patients, one in the liberal-GC group and one in the tight-GC group, were not contactable 90 days after randomization.

Figure 3: Forest plots visualizing tight-GC versus liberal-GC in subgroups. The widths of the 95% confidence intervals (CI) have not been adjusted for multiplicity and may not be used in place of hypothesis testing. Interaction analysis revealed possible heterogeneity in treatment effect for patients admitted with a neurological/neurosurgical admission diagnosis.

Table 1. Baseline Characteristics

Characteristic*	Liberal glucose control (N=4622)	Tight glucose control (N=4608)
Age - yr	67 (56-75)	67 (57-75)
Male sex - no. (%)	2902 (62.8%)	2930 (63.6%)
Weight - kg	75 (65-86)	76 (65-88)
Body-mass index †	26 (23-29)	26 (23-29)
History of diabetes mellitus - no. (%)	955 (20.7%)	933 (20.2%)
Charlson comorbidity index ‡	4 (2-6)	4 (2-5)
Sepsis upon admission - no. (%) §	1307 (28.3%)	1313 (28.5%)
APACHE II score upon admission ¶	21 (15-30)	21 (15-30)
Predicted ICU stay >1 week - no. (%)	1605 (34.7%)	1625 (35.3%)
Diagnostic category at ICU admission - no. (%)		
Cardiac surgery	2084 (45.1%)	2086 (45.3%)
Neurological/Neurosurgical	528 (11.4%)	525 (11.4%)
Complicated abdominal/pelvic surgery	522 (11.3%)	505 (11.0%)
Solid organ transplantation	319 (6.9%)	314 (6.8%)
Complicated lung/esophageal surgery	168 (3.6%)	179 (3.9%)
Trauma/burns	172 (3.7%)	175 (3.8%)
Gastrointestinal/hepatic	151 (3.3%)	158 (3.4%)
Respiratory	139 (3.0%)	137 (3.0%)
Complicated vascular surgery	117 (2.5%)	117 (2.5%)
Cardiovascular	106 (2.3%)	103 (2.2%)
Hematological/oncological	70 (1.5%)	65 (1.4%)
Metabolic/Renal	40 (0.9%)	40 (0.9%)
Other	206 (4.5%)	204 (4.4%)
Admission blood glucose - mg/dL $	143 (120-170)	142 (121-168)

* Continuous data are presented as medians (25th-75th percentile). Abbreviations: APACHE II denotes Acute Physiology And Chronic Health Evaluation II; ICU denotes intensive care unit.

† The body-mass index is the weight in kilograms divided by the square of the height in meters. Data from 4 patients are missing.
‡ Scores on the Charlson comorbidity index range from 0 to 37, with higher scores indicating more comorbidity.[33]
§ Sepsis was scored according to the Sepsis-3 definition.[34]
¶ The scoring of baseline illness severity was based on APACHE II score, ranging from 0 to 71, with higher scores indicating a greater baseline illness severity.[35]
$ To convert blood glucose in mg/dL to mmol/L, divide by 18.

Table 2. Blood Glucose Management, Nutrition and Outcomes

Variable*	Liberal glucose control (N=4622)	Tight glucose control (N=4608)	Relative risk (95% CI) or Difference (95% CI)	P value
Protocol compliance: glucose metrics and nutrition				
Treated with insulin - no. (%)	2122 (45.9%)	4551 (98.8%)	2.15 (2.08-2.22)	
Average daily insulin dose - units/day	0.0 (0.0-5.6)	24.8 (14.8-39.9)	21.0 (20.5-21.5)	
Peak blood glucose in ICU - mg/dL	189 (162-230)	170 (146-204)	-19 (-21 to -17)	
Average daily blood glucose - mg/dL	145 (128-165)	115 (107-126)	-28 (-29 to -27)	
Average morning blood glucose - mg/dL †	140 (122-161)	107 (98-117)	-32 (-33 to -32)	
Hypoglycemia (<40 mg/dL) - no. (%)	31 (0.7%)	47 (1.0%)	1.52 (0.97-2.39)	
Parenteral nutrition (average kcal/day from day 1 to day 7)	101 (77-160)	101 (76-162)	0 (-2 to 2)	
Primary and safety endpoints				
Time to discharge alive from ICU - Hazard ratio (95% CI)		1.00 (0.96-1.04)		0.94
Duration of stay in ICU - days	7±13 3 (1-6)	6±12 3 (1-6)	0 (-1 to 0) 0 (0-0)	
Mortality within 90 days - no. (%)	468/4621 (10.1%)	486/4607 (10.5%)	1.04 (0.92-1.17)	0.51
Secondary outcomes				
Acute kidney injury ‡				
AKI-3 - no. (%)	387/4500 (8.6%)	326/4507 (7.2%)	0.84 (0.73-0.97)	
New RRT - no. (%)	259/4500 (5.8%)	212/4507 (4.7%)	0.82 (0.68-0.98)	
Liver dysfunction				
Peak gamma GT ≥90 U/L - no. (%) §	1370/4170 (32.9%)	1175/4146 (28.3%)	0.86 (0.81-0.92)	
Peak alkaline phosphatase ≥195 U/L - no. (%) §	498/3818 (13.0%)	437/3787 (11.5%)	0.88 (0.78-0.99)	
Peak bilirubin >3mg/dL - no. (%)	458/4545 (10.1%)	409/4526 (9.0%)	0.90 (0.79-1.02)	
Peak AST ≥111 U/L - no. (%) ¶	1013/4584 (22.1%)	1020/4578 (22.3%)	1.01 (0.93-1.09)	
Peak ALT ≥123 U/L - no. (%) ¶	723/4583 (15.8%)	677/4574 (14.8%)	0.94 (0.85-1.03)	
Infection				
Patients with a new infection - no. (%)	664 (14.4%)	634 (13.8%)	0.96 (0.87-1.06)	
Respiratory support				
Time to live weaning from mechanical respiratory support - Hazard ratio (95% CI)		0.99 (0.95-1.03)		

Hemodynamic support				
Time to live weaning from hemodynamic support - Hazard ratio (95% CI)		0.98 (0.94-1.02)		
Time in hospital				
Time to discharge alive from hospital - Hazard ratio (95% CI)		1.00 (0.96-1.04)		
Mortality in ICU - no. (%)	247 (5.3%)	267 (5.8%)	1.08 (0.92-1.28)	
Mortality in hospital - no. (%)	421 (9.1%)	443 (9.6%)	1.06 (0.93-1.20)	

* Continuous data are presented as medians (25th-75th percentile), unless reported as plus-minus values, which denote means±SD. To convert blood glucose in mg/dL to mmol/L, divide by 18. For continuous variables, differences between the groups are presented as differences in medians (Hodges-Lehman estimate) or differences in means; for proportions, differences between the groups are presented as relative risk. The widths of the 95% CI have not been adjusted for multiplicity and may not be used in place of hypothesis testing. Abbreviations: AKI denotes acute kidney injury; ALT denotes alanine transaminase; AST denotes aspartate transaminase; CI denotes confidence intervals; ICU denotes intensive care unit; gamma GT denotes gamma glutamyltransferase; RRT denotes renal replacement therapy.
† Data missing in 91 patients.
‡ To study the incidence of AKI-3 and the use of new RRT, patients on chronic dialysis or having received RRT before randomization were excluded (N=223).
§ Minimum 1.5 times the upper limit of normality.
¶ Minimum 3 times the upper limit of normality.